Code number:	TESLA	Description:	gen. communication specification
Project number:	D531601	Customer:	Tesla

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General Diagnostic Communication

Tesla

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Issue

No.	Date	Chapter	Description	Author
0.1	21.04.10	all	Initial version	MKR
0.2	23.04.10	6	Add chapter 6 for DTC status byte	IPO
0.3	29.04.10	5.3	Update chapter with ALI-Service	IPO
0.4	27.08.10	5.2	adapted environmental Data	MKR
0.5	18.02.11	5.4	update environmental data information	MKR
0.6	01.03.11	5.2	remove data block A001-A007,add additional default data blocks A045, A047	MKR
0.7	03.03.11	5.1 5.2	add diagnostic services add default data blocks	MKR

Table 1: Issue

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1 Introduction

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This product specification applies to the ECU for 2-axle air suspension system with closed air control supply.

Drawing data have priority over specification data.

2 Definitions

2.1 Abbreviations

CAN	Controller Area Network
CCU	Chassis Control Unit
DC	Damping Control
EEPROM	Electrically Erasable Programmable Read Only Memory
EAS	Electronic Air Suspension
DTC	Diagnostic Trouble Code

Table 2

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3 Referenced Documents

The CAN implementation of the Chassis Control Unit project is based on the reference documents shown below.

No.	title	Document	Version	Date
1	Road vehicles Diagnostics on Controller Area Networks (CAN) Part 2:Network layer service	ISO 15765-2		
2	Road vehicles Diagnostics on Controller Area Networks(CAN) Part3:Implementation of unified diagnostic services (UDS on CAN)	ISO 15765-3		
3	Road vehicles Unified diagnostic services (UDS)	ISO 14229		
4	Diagnostic Trouble Code Definitions Equivalent to ISO/DIS 15031-6:April 30, 2002	SAE J2012	REV. APR2002	
5	Controller Area Network (CAN) Part 1 Data Link layer and physical signalling	ISO 11898-1		
6	Controller Area Network (CAN) Part 2 High-speed medium access unit	ISO 11898-2		

Table 3

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4 Standard CAN

4.1 Diagnostic Parameters

Name	Category	Value	Description
CAN Bus Class	Can Bus Class C		compliant to ISO 11898
CAN Baudrate	Communication	500000	Used baud rate for communication with the ecu.
FunctionalCANIdentifierSessionHandling	Communication	0x441	Can identifier for functional initialization, tester present and diagnostic end.
FunctionalCANIdentifier	Communication	0x441	Can identifier for functional requests.
PhysicalRequestCANIdentifier	Communication	0x64B	Used CAN identifier for requests.
PhysicalResponseCANIdentifier	Communication	0x65B	Used CAN identifier for responses.
UsedSTminInTesterFC	Communication	20 [ms]	Expected cycle time for consecutive frames of a response. Transmitted in STmin of the tester flow control.
UsedBlockSizeInTesterFC	Communication	0x08	Describes how many can messages are to be received between two (2) flow control frames. Corresponds to the Block Size included in the flow control frame of the tester.
P2Timeout	Communication	150 [ms]	Maximum time between request and response.
P2ExtendedTimeout	Communication	2000 [ms]	Within this time a ecu has to send a response after it has requested more time via RC 7F 78.
UsedRequestTimeForPhysicalTesterPresent	Communication	2000 [ms]	Time between a response and the next subsequent tester present message (if no other request is sent to this ECU) in case of physically addressed requests.
UsedRequestTimeForFunctionalTesterPresent	Communication	2000 [ms]	Time between two periodically transmitted tester present requests (functional request without positive response handling).

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Name	Category	Value	Description
Busy/PendingCounter	Communication	30	Number of repetitions for negative response code BusyRepeatRequest (7F xx 21), ResponsePending (7F xx 78). The COM parameter is read again from the variant buffer before each repetition and if the result is lower than the internal value then the new value is used. So it is possible to stop the repetition by the application. The repetition stops if the value becomes 0.
Address mode	Communication	AM_NORMAL	Address mode for ISO 15765 (AM_NORMAL: 11 bit identifier)

Table 4

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5 Diagnostic Service Support

5.1 Diagnostic Services

Diagnostic Session Control [$10]
Supported Diagnostic Sessions are as follows:
Default Session [$01/$81]
Programming Session [$02/$82]
Extended Diagnostic Session [$03/$83]

ECU Reset ($11)
Supported Reset Sessions:
Hard Reset [$01/$81]
Soft Reset [$03/$83]
Enable Rapid Power Shutdown [$04/$84]

Clear Diagnostic Information ($14)

Read DTC Information [$19]
Report Number Of DTC By Status Mask [$01]
Report DTC By Status Mask [$02]
Report DTC Extended Data Record By DTC Number [$06]
Report Supported DTCs [$0A]
Report Report Mirror Memory DTC By StatusMask [$0F]
Report Mirror Memory DTC Extended Data By DTC Nbr Fault Memory [$010]
Report Nbr Of Mirror Memory DTC By Status Mask Fault Memory [$11]
Report DTC Fault Detection Counter [$14]

Read/Write Data By Identifier [$22/3E]
Read Data By Identifier [$22]
Write Data By Identifier [$2E]

see chapter 5.2

Security Access ($27)

Communication Control [$28]
Enable Rx and Tx [$00/$80]
Enable Rx and Disable Tx [$01/$81]

Routines ($31)

see chapter 5.3

Draft

Request Download ($34)

Transfer Data ($36)

Request Transfer Exit ($37)

Tester Present [$3E]
Response Required ($00)
No Response Required ($80)

Control DTC [$85]
DTC Setting Type (On) [$01/$81]
DTC Setting Type (Off) [$02/$82] - no DTC's will be stored into memory

Table 5

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5.2 Read Write Data By Data Identifier Summary [$22/2E]

Read Write Data By Data Identifier Summary

SID	Data ID	Data Identifier Description	Function	# of Bytes Request	# of Bytes Response	Security Level	Engineering	Manufacturing	Service
$22	$0100	Reprogramming Attempt Counter	Read	3	15		yes	no	no
$22	$010E	Re-Programming Error Status	Read	3	4		yes	no	yes
$22	$A040	Data Block tbd	Read	3			yes	yes	yes
$22	$A041	Data Block tbd	Read	3			yes	yes	yes
$22	$A042	Data Block tbd	Read	3			yes	yes	yes
$22	$A043	Data Block tbd	Read	3			yes	yes	yes
$22	$A044	Data Block tbd	Read	3			yes	yes	yes
$22	$A045	Data Block tbd	Read	3			yes	yes	yes
$22	$A046	Data Block tbd	Read	3			yes	yes	yes
$22	$A047	Data Block tbd	Read	3			yes	yes	yes
$22	$A048	Data Block tbd	Read	3			yes	yes	yes
$22	$A049	Data Block tbd	Read	3			yes	yes	yes
$22	$A04A	Data Block tbd	Read	3			yes	yes	yes
$22	$A030	FL Calibrated Height	Read	3	5		yes	yes	yes
$2E	$A030	FL Calibrated Height	Write	5	3		yes	yes	yes
$22	$A031	FR Calibrated Height	Read	3	5		yes	yes	yes
$2E	$A031	FR Calibrated Height	Write	5	3		yes	yes	yes
$22	$A032	RL Calibrated Height	Read	3	5		yes	yes	yes
$2E	$A032	RL Calibrated Height	Write	5	3		yes	yes	yes
$22	$A033	RR Calibrated Height	Read	3	5		yes	yes	yes
$2E	$A033	RR Calibrated Height	Write	5	3		yes	yes	yes
$22	$A034	Read Stored Calibration Values Permanent Status	Read	3	4		yes	yes	yes
$2E	$A034	Read Stored Calibration Values Permanent Status	Write	4	3		yes	yes	no
$22	$F100	Active Diagnostic Information	Read	3	7		yes	no	yes
$22	$F112	Customer Hardware Part Number	Read	3	13		yes	no	yes
$22	$F122	Customer Software Part Number	Read	3	13		yes	no	yes
$22	$F132	ECU Part Number	Read	3	13		yes	no	yes
$2E	$F132	ECU Part Number	Write	13	3		yes	no	yes
$22	$F150	Hardware Version Information	Read	3	6		yes	no	yes
$22	$F151	Software Version Information	Read	3	6		yes	no	yes

$22	$F153	Boot Software Version Information	Read	3	6		yes	no	yes
$22	$F154	Hardware Supplier Identification	Read	3	5		yes	no	yes
$22	$F155	Software Supplier Identification	Read	3	5		yes	no	yes
$22	$F18C	ECU Serial Number	Read	3	17		yes	no	yes

Table 6

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5.3 Routine By Data Identifier Summary [$31]

Routine By Data Identifier Summary

SID	Param	Data ID	Data Identifier Description	Function	# of Bytes Request	# of Bytes Response	Security Level	Engineering	Manufacturing	Service
$31	$01/$81	$0305	Leveling to set level (ALI)	Start Routine	5	4		yes	yes	no
$31	$02/$82	$0305	Leveling to set level (ALI)	Stop Routine	4	4		yes	yes	no
$31	$03	$0305	Leveling to set level (ALI)	Return Results	4	5		yes	yes	no
$31	$01/$81	$0306	Enable/Disable Level Control (ALI)	Start Routine	5	4		yes	yes	no
$31	$02/$82	$0306	Enable/Disable Level Control (ALI)	Stop Routine	4	4		yes	yes	no
$31	$03	$0306	Enable/Disable Level Control (ALI)	Return Results	4	5		yes	yes	no
$31	$01/$81	$0309	Finish Assembly Line Initialization (ALI)	Start Routine	4	4		yes	yes	no
$31	$02/$82	$0309	Finish Assembly Line Initialization (ALI)	Stop Routine	4	4		yes	yes	no
$31	$03	$0309	Finish Assembly Line Initialization (ALI)	Return Results	4	5		yes	yes	no
$31	$01/$81	$030A	Measuring System Airmass (ALI)	Start Routine	4	4		yes	yes	no
$31	$02/$82	$030A	Measuring System Airmass (ALI)	Stop Routine	4	4		yes	yes	no
$31	$03	$030A	Measuring System Airmass (ALI)	Return Results	4	8		yes	yes	no
$31	$01/$81	$030B	Actuator Test (ALI)	Start Routine	5	4		yes	yes	no
$31	$02/$82	$030B	Actuator Test (ALI)	Stop Routine	4	4		yes	yes	no
$31	$03	$030B	Actuator Test (ALI)	Return Results	4	5		yes	yes	no
$31	$01/$81	$030C	Height Sensor Check (ALI)	Start Routine	4	4		yes	yes	no
$31	$02/$82	$030C	Height Sensor Check (ALI)	Stop Routine	4	4		yes	yes	no
$31	$03	$030C	Height Sensor Check (ALI)	Return Results	4	5		yes	yes	no
$31	$01/$81	$0310	Ambience Valve Actuator Test (ALI)	Start Routine	4	4		yes	yes	no
$31	$02/$82	$0310	Ambience Valve Actuator Test (ALI)	Stop Routine	4	4		yes	yes	no
$31	$03	$0310	Ambience Valve Actuator Test (ALI)	Return Results	4	5		yes	yes	no
$31	$01/$81	$0314	Component Pressure Measurement	Start Routine	5	4		yes	yes	yes
$31	$02/$82	$0314	Component Pressure Measurement	Stop Routine	4	4		yes	yes	yes

$31	$03	$0315	Component Pressure Measurement all	Return Results	4	6		yes	yes	yes
$31	$01/$81	$0315	Component Pressure Measurement all	Start Routine	5	4		yes	yes	yes
$31	$02/$82	$0315	Component Pressure Measurement all	Stop Routine	4	4		yes	yes	yes
$31	$03	$0315	Component Pressure Measurement all	Return Results	4	11		yes	yes	yes
$31	$01/$81	$0332	Air Filling via Reservior (Service)	Start Routine	6	4		yes	no	yes
$31	$02/$82	$0332	Air Filling via Reservior (Service)	Stop Routine	4	4		yes	no	yes
$31	$03	$0332	Air Filling via Reservior (Service)	Return Results	4	5		yes	no	yes
$31	$01/$81	$0333	Deflating Airspring into Reservior (service)	Start Routine	6	4		yes	no	yes
$31	$02/$82	$0333	Deflating Airspring into Reservior (service)	Stop Routine	4	4		yes	no	yes
$31	$03	$0333	Deflating Airspring into Reservior (service)	Return Results	4	5		yes	no	yes
$31	$01/$81	$0334	Deflating Component into Ambience	Start Routine	6	4		yes	no	yes
$31	$02/$82	$0334	Deflating Component into Ambience	Stop Routine	4	4		yes	no	yes
$31	$03	$0334	Deflating Component into Ambience	Return Results	4	5		yes	no	yes
$31	$01/$81	$0351	Simple Time-Controlled System Fill	Start Routine	6	4		yes	yes	yes
$31	$02/$82	$0351	Simple Time-Controlled System Fill	Stop Routine	4	4		yes	yes	yes
$31	$03	$0351	Simple Time-Controlled System Fill	Return Results	4	5		yes	yes	yes

Table 7

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5.4 ENVIRONMENTAL DATA SET 100

Response of Read DTC request. Conversion definitions: see Table 9 and Table 10.

ENV Data Set 100

Byte [Bit]	Byte / Bit Description	Default Value	Size	Conversion Type	Conversion Name	Conversion ID
0 [0]	Read DTC Information Response	$59	1 [0]	-	-	-
1 [0]	Report DTC Extended Data By DTC Number	$06	1 [0]	Identical	UDS parameters -- constant/literal values	3
2 [0]	DTC		3 [0]	Identical	UDS parameters	9
5 [0]	Test Failed		0 [1]	Table	Status of DTC - Bit Definition	4
5 [1]	Test Failed This Operation Cycle		0 [1]	Table	Status of DTC - Bit Definition	4
5 [2]	Pending DTC		0 [1]	Table	Status of DTC - Bit Definition	4
5 [3]	Confirmed DTC		0 [1]	Table	Status of DTC - Bit Definition	4
5 [4]	Test Not Completed Since Last Clear		0 [1]	Table	Status of DTC - Bit Definition	4
5 [5]	Test Failed Since Last Clear		0 [1]	Table	Status of DTC - Bit Definition	4
5 [6]	Test Not Completed This Operation Cycle		0 [1]	Table	Status of DTC - Bit Definition	4
5 [7]	Warning Indicator Requested		0 [1]	Table	Status of DTC - Bit Definition	4
6 [0]	**DTC Extended Data Record Number - All**		1 [0]	Identical	UDS parameters -- constant/literal values	3
7 [0]	Occurrence Flag		0 [1]	Table	Error/Occurrence (1Bit)	1744
7 [1]	Reserved		0 [7]	Table	Reserved Special Conversion	1755
8 [0]	Original Odometer Value		2 [0]	Linear	OdometerValue (2 Byte)	1377
10 [0]	Most Recent Odometer Value		2 [0]	Linear	OdometerValue (2 Byte)	1377
12 [0]	Frequency Counter		1 [0]	Linear	Frequency Counter (1Byte)	1406
13 [0]	Ignition Cycle Counter		1 [0]	Linear	Ignition Cycle Counter (1Byte) PLinear	2775
14 [0]	**DTCExtendedDataRecordNumber 2**		1 [0]	Linear	Unsigned (1Byte)	2766
15 [0]	System Voltage		1 [0]	Linear	0 to 25.5 V (0.1, 0)	156
16 [0]	Vehicle Speed		2 [0]	Linear	0 to 921.4221 km/h (.0141, 0, 5)	4315
18 [0]	Engine RPM		1 [0]	Linear	0 to 5100 rpm (20, 0, 0)	4558
19 [0]	Compressor Temperature		1 [0]	Linear	-40 to 214 C (1, -40, 0)	4559
20 [0]	Reverse Valve 1		0 [1]	Table	off/on	5363
20 [1]	Reverse Valve 2		0 [1]	Table	off/on	5363
20 [3]	Reverse Valve High-Side		0 [1]	Table	off/on	5363
20 [4]	Ignition Run/Start		0 [1]	Table	off/on	5363
21 [0]	FL Level		1 [0]	Linear	-100 to 153 mm (1, -100, 0)	4562

22 [0]	FR Level		1 [0]	Linear	-100 to 153 mm (1, -100, 0)	4562
23 [0]	RL Level		1 [0]	Linear	-100 to 153 mm (1, -100, 0)	4562
24 [0]	RR Level		1 [0]	Linear	-100 to 153 mm (1, -100, 0)	4562
25 [0]	System Pressure		1 [0]	Linear	0 to 35 Bar (.13725, 0, 5)	4310
26 [0]	FL Control Level Valve		0 [1]	Table	off/on	5363
26 [1]	FR Control Level Valve		0 [1]	Table	off/on	5363
26 [2]	RL Control Level Valve		0 [1]	Table	off/on	5363
26 [3]	RR Control Level Valve		0 [1]	Table	off/on	5363
26 [4]	Valve Control High-Side		0 [1]	Table	off/on	5363
26 [5]	Ambient -Valve		0 [1]	Table	off/on	5363
26 [6]	Control Compressor relay		0 [1]	Table	off/on	5363
26 [7]	Compressor Relay High-Side		0 [1]	Table	off/on	5363
27 [0]	Current Level Status		0 [4]	Table	Current Level Status	7950
27 [4]	Target Level Status		0 [4]	Table	Target Level Status	7953
28 [0]	Service Air Suspension, See Dealer now (Red)		0 [1]	Table	No/Yes	1990
28 [1]	Driver Door Ajar		0 [1]	Table	Closed/Open 1 Bit	6933
28 [2]	Passenger Door Ajar		0 [1]	Table	Closed/Open 1 Bit	6933
28 [3]	Right Rear Door Ajar		0 [1]	Table	Closed/Open 1 Bit	6933
28 [4]	Left Rear Door Ajar		0 [1]	Table	Closed/Open 1 Bit	6933
28 [5]	Rear Trunk Ajar		0 [1]	Table	Closed/Open 1 Bit	6933
28 [6]	Front Trunk Ajar		0 [1]	Table	Closed/Open 1 Bit	6933

Table 8: environmental data structure

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5.5 Linear Conversion Table (ID)

Linear Conversions

ID	Description	Size	y(x)=m(x)+b Factor (m)	Offset (b)	Type	Decimal Places	Units	System	Scaled Minimum	Scaled Maximum	State Encodings (MIN/MAX - Meaning)
111	0 to 8191 rpm (0.5, 0)	2 [0]	0,5	0	UNUM	1	rpm	OTHER	0	8191	($FFFF/$FFFF - SNA)
146	0 to 12282 Nm (3,0)	2 [0]	3	0	UNUM	2	N*m	SI	0	12282	($0FFF/$0FFF - SNA)
148	0 to 65534 rpm (1, 0)	2 [0]	1	0	UNUM	0	rpm	OTHER	0	65534	($FFFF/$FFFF - SNA)
156	0 to 25.5 V (0.1, 0)	1 [0]	0,1	0	UNUM	1	Volts	OTHER	0	25,5	
168	0 to 255 mm	1 [0]	1	0	UNUM	0	mm	SI	0	255	
170	-40 to 85 C (0.5, -40)	1 [0]	0,5	-40	UNUM	1	C	SI	-40	85	($FF/$FF - SNA / Not Programmed)
194	0 to 255 Sec (1,0)	1 [0]	1	0	UNUM	0	seconds	OTHER	0	255	
236	-40 to 85 C (0.01, -40)	2 [0]	0,01	-40	UNUM	2	C	SI	-40	85	($FFFF/$FFFF - SNA)
286	-10.24 to 10.08 meters/sec^2 (0.08, -10.24)	1 [0]	0,08	-10,24	UNUM	2	m/s^2	SI	-10,24	10,08	($FF/$FF - SNA / Not Programmed)
377	-2048 to +2047 degrees (0.5,0)	2 [0]	0,5	-2048	UNUM	1	degrees	OTHER	-2048	2047	($3FFF/$3FFF - SNA)
691	0 to 65535 mm (1,0,0)	2 [0]	1	0	UNUM	0	mm	SI	0	65535	
915	0 to 255 % (1,0,0)	1 [0]	1	0	UNUM	0	%	OTHER	0	255	
1173	0 to 254 kPa (1,0)	1 [0]	1	0	UNUM	0	kPa	SI	0	254	($FF/$FF - SNA / Not Programmed)
1207	Unsigned (2Byte)	2 [0]	1	0	UNUM	0		SI	0	65535	

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1353	0 to 63.5 N*m (0.25,0) w/SNA	1 [0]	0,25	0	UNUM	2	N*m	SI	0	63,5	($FF/$FF - SNA / Not Programmed)
1359	0 to 1.98[No Units] (.0077953, 0)	1 [0]	0,007795	0	UNUM	4		OTHER	0	1,9800062	
1367	Unsigned (4Byte)	4 [0]	1	0	UNUM	0		SI	0	4,295E+09	
1377	OdometerValue (2 Byte)	2 [0]	16	0	UNUM	0	km	SI	0	1048544	($FFFF/$FFFF - Odometer value not available / Default)
1392	Osciallator Frequency (2Byte)	2 [0]	0,001	0	UNUM	3	MHz	SI	0	65,535	
1406	Frequency Counter (1Byte)	1 [0]	1	0	UNUM	0		SI	1	255	($00/$00 - Frequency counter not available / Default)
1413	Resistor Value (2Byte)	2 [0]	0,001	0	UNUM	3	kOhm	SI	0	65,535	
2667	Month	1 [0]	1	0	UNUM	0		SI	1	12	($00/$00 - Undefined) , ($0D/$FF - Undefined)
2673	Day	1 [0]	1	0	UNUM	0		SI	1	31	($00/$00 - Undefined) , ($20/$FF - Undefined)
2684	NetworkID	1 [0]	1	0	UNUM	0		SI	0	255	
2766	Unsigned (1Byte)	1 [0]	1	0	UNUM	0		SI	0	255	
2775	Ignition Cycle Counter (1Byte) PLinear	1 [0]	1	0	UNUM	0		SI	0	255	
2832	Week (1Byte)	1 [0]	1	0	UNUM	0		SI	1	52	($00/$00 - Invalid value) , ($35/$FF - Invalid value)
3167	-500 to 1547.5 N*m (0.25,-500,2)	2 [0]	0,25	-500	UNUM	2	N*m	SI	-500	1547,5	($1FFF/$1FFF - SNA)
3183	-40 to 214 C (1, -40)	1 [0]	1	-40	UNUM	0	DegC	SI	-40	214	($FF/$FF - SNA)
4309	0 to 5 V (.01961, 0, 5)	1 [0]	0,01961	0	UNUM	5	volt	SI	0	5,00055	
4310	0 to 35 Bar (.13725, 0, 5)	1 [0]	0,13725	0	UNUM	5	bar	SI	0	34,99875	
4315	0 to 921.4221 km/h (.0141, 0, 5)	2 [0]	0,01406	0	UNUM	5	km/h	SI	0	921,4221	
4558	0 to 5100 rpm (20, 0, 0)	1 [0]	20	0	UNUM	0	RPM	SI	0	5100	
4559	-40 to 214 C (1, -40, 0)	1 [0]	1	-40	UNUM	0	C	SI	-40	214	

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4562	-100 to 153 mm (1, -100, 0)	1 [0]	1	-100	UNUM	0	mm	SI	-100	153	($FF/$FF - SNA) , ($FE/$FE - Not Init)
4780	2000 to 2062 (1,2000)	1 [0]	1	2000	UNUM	0	No Unit	OTHER	2000	2062	($FF/$FF - Not Programmed)
4830	0 to 255 No Units (1,0,0)	1 [0]	1	0	UNUM	0	No Units	SI	0	255	
5071	-327.68 to 327.67 mm (0.01,0,0)	2 [0]	0,01	-327,68	SNUM	0	mm	SI	-327,68	327,66	
5072	0 to 508 km/h (2, 0, 0)	1 [0]	2	0	UNUM	0	Km/h	SI	0	508	($FF/$FF - SNA / Not Programmed)
5073	-327.68 to +327.66 Deg c/sec (0.01, 0, 2)	2 [0]	0,01	-327,68	SNUM	2	Deg traveled /sec	OTHER	-327,68	327,66	($FFFF/$FFFF - SNA)
5074	0 to 100 % (0.4, 0, 1)	1 [0]	0,4	0	UNUM	1	%	OTHER	0	100	($FF/$FF - SNA / Not Programmed)
5075	0 to 25.5 Bars (0.1, 0, 1)	1 [0]	0,1	0	UNUM	1	Bar	SI	0	25,5	
5153	0 to 6553.5 Bar-liter (0.1, 0, 0)	2 [0]	0,1	0	UNUM	0	Bar-liter	SI	0	6553,5	
NEW	OdometerValue (3 Byte)	3 [0]	0,1	0	UNUM	1	km	SI	0	1677721,4	($FFFFFF/$FFFFFF - Odometer value not available / Default)

Table 9: linear conversion overview

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5.6 Identical Conversions (ID)

Identical Conversions

ID	Name	Description	Size	Type
1	KWP SID conversion and UDS SID conversion		1 [0]	sHEX
3	UDS parameters -- constant/literal values		1 [0]	sHEX
5	KWP 2 byte LIDs and UDS DIDs		2 [0]	sHEX
9	UDS parameters		3 [0]	sHEX
160	HexDump (2Byte)		2 [0]	sHEX
162	HexDump (4Byte)		4 [0]	sHEX
163	Bcd (1Byte)		1 [0]	sBCD
165	HexDump (1Byte)		1 [0]	sHEX
198	HexDump (3Byte)		3 [0]	sHEX
201	Start		1 [0]	sHEX
202	Stop		1 [0]	sHEX
220	ASCII (14Byte)		14 [0]	sASC
340	VIN (17Byte)		17 [0]	sASC
342	ASCII (10Byte)		10 [0]	sASC

Table 10:identical conversion overview

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6 Fault Management

6.1 Status of DTC

This section defines the DTC Status Information Byte representing the maturation and storage state of an individual DTC. All Status Bits for Non-Emission Related ECU's are supported.

Bit	Description	ECU Support Mandatory / Optional	
		Emission Related	Non-Emission Related
0	**Test Failed** This bit shall indicate the result of the most recently performed test. A logical '1' shall indicate that the last test failed meaning that the failure is completely matured (Active). A logical '0' shall indicate that the result of the most recently performed test returns a "pass" result. 1: most recent result from DTC test indicated a failed result. 0: most recent result from DTC test indicated no failure detected.	M	M
1	**Test Failed This Operation Cycle** This bit shall indicate either that a diagnostic test has reported a Test Failed result at any time during the current operation cycle. Reset to 0 when a new operation cycle is initiated or after a call to Clear Diagnostic Information. Once this bit is set to 1, it shall remain a 1 until a new operation cycle is started. 1: Test Failed bit = 1 has failed at least once during the current operation cycle. 0: Test Failed bit = 0 has not failed during the current operation cycle or after a call was made to Clear Diagnostic Information during the current operation cycle.	M	O
2	**Pending DTC** This bit shall indicate whether or not a diagnostic test has reported a Test Failed result at any time during the current or last completed operation cycle. The status shall only be updated if the test runs and completes. The criteria to set the Pending DTC bit and the Test Failed This Operation Cycle bit are the same. The difference is that the Test Failed This Operation Cycle is cleared at the end of the current operation cycle and the Pending DTC bit is not cleared until an operation cycle has completed where the test has passed at least once and never failed. If the test did not complete during the current operation cycle, the status bit shall not be changed. For example, if a monitor stops running after a confirmed DTC is set, the Pending DTC must remain set = '1'. For an OBD DTC, a pending DTC is required to be stored after a malfunction is detected during the first driving cycle. 1: This bit shall be set to 1 and latched if a malfunction is detected during the current operation cycle. 0: This bit shall be set to 0 after completing an operation cycle during which the test completed and a malfunction was not detected or upon a call to the Clear Diagnostic Information service.	M	O

Bit	Description	ECU Support Mandatory / Optional	
		Emission Related	Non-Emission Related
3	**Confirmed DTC** This bit shall indicate whether a malfunction was detected enough times to warrant that the DTC is stored in long-term memory (Pending DTC has been set = '1' one or more times for emission relevant electronic control units). This information can be used by the external test tool to request additional diagnostic information such as Extended Data Records or Snapshot Records. A Confirmed DTC does not indicate that the malfunction is present at the time of the request (Pending DTC or Test Failed can be used to determine if a malfunction is present at the time of the request.). Reset to logical '0' after a call to Clear Diagnostic Information or after self-healing criteria have been satisfied (e.g. 40 engine warm-up cycles without another detected malfunction for emissions-related ECUs, 100 ignition cycles without another detected malfunction for all other ECUs). Furthermore this bit is reset when the fault record associated with this DTC is overwritten by a newer DTC with higher priority due to a fault memory overflow. 1 : DTC has failed at least once since the last call to Clear Diagnostic Information and self-healing criteria have not yet been satisfied and the DTC has not been removed from the fault memory due to an overflow of the available fault records 0: DTC has not failed since the last call to Clear Diagnostic Information or after the selfhealing criteria have been satisfied for the DTC	M	M
4	**Test Not Completed Since Last Clear** This bit shall indicate whether a DTC test has run to completion since the last time a call was made to Clear Diagnostic Information. One (1) shall indicate that the DTC test has not run to completion. If the test runs and passes or fails (Test Failed This Operation Cycle = 1) then the bit shall be set to a Zero (0) and latched. Reset to One (1) after a call to Clear Diagnostic Information. 1: DTC test has not run to completion since the last time diagnostic information was cleared. 0: DTC test has returned either a passed or failed test result at least one time since the last time diagnostic information was cleared.	M	M

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Bit	Description	ECU Support Mandatory / Optional	
		Emission Related	Non-Emission Related
5	**Test Failed Since Last Clear** This bit shall indicate whether a DTC test has ever returned a Test Failed = 1 result since the last time a call was made to Clear Diagnostic Information (latched Test Failed This Monitoring Cycle =1). Zero (0) shall indicate that the test has not run or that the DTC test ran and passed (but never failed). If the test runs and fails then the bit shall remain latched at a 1.Reset to Zero (0) after a call to Clear Diagnostic Information. In contradiction to the Confirmed DTC this bit is not reset by self-healing criteria or when it was overwritten due to an overflow of the fault memory. 1: DTC test returned a Test Failed This Operation Cycle = 1 result at least once since the last time diagnostic information was cleared. 0: DTC test has not indicated a Test Failed This Operation Cycle = 1 result since the last time diagnostic information was cleared.	M	O
6	**Test Not Completed This Operation Cycle** This bit shall indicate whether a DTC test has ever run and completed during the current operation cycle. One (1) shall indicate that the DTC test has not run to completion during the current operation cycle. If the test runs and passes or fails then the bit shall be set (and latched) to 0 until a new operation cycle is started. Reset to 1 after a call to Clear Diagnostic Information. 1: DTC test has not run to completion this operation cycle (or since the last time diagnostic information was cleared this operation cycle). 0: DTC test has returned either a passed or Test Failed This Operation Cycle = 1 result during the current operation cycle or a call was made to Clear Diagnostic Information.	M	O

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Bit	Description	ECU Support Mandatory / Optional	
		Emission Related	Non-Emission Related
7	**Warning Indicator Requested** This bit shall report the status of any warning indicators associated with a particular DTC. Warning outputs may consist of indicator lamp(s), displayed text information, etc. If no warning indicators exist for a given system or particular DTC, this status shall default to a logic"0" state. Conditions for activating and deactivating the warning indicator are defined by the application specific document. Reset to a logical '0' after a call to Clear Diagnostic Information. 1: Warning indicator requested to be ON. 0: Warning indicator requested to be OFF. If the warning indicator is on for a given DTC, then Confirmed DTC shall be also be set to 1.	M	O

Table 9: Status of DTC Bit Definition

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